

A Quantum Threat Model for Hyperledger Fabric: Cryptographic Audit, Risk Prioritisation, and Formal Integration Requirements

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Abstract—Cryptographically relevant quantum computers (CRQCs) threaten every system whose security rests on the elliptic curve discrete logarithm problem (ECDLP). Hyperledger Fabric, the dominant enterprise permissioned blockchain, depends uniformly on ECDSA and ECDH over P-256 across its entire transaction lifecycle and identity infrastructure. This paper presents the first component-level quantum vulnerability audit and formal threat model for Hyperledger Fabric. Auditing v2.5.12, we identify twenty-five distinct quantum-vulnerable operations across five lifecycle phases, the Fabric CA and MSP identity subsystems, the gossip protocol, and the key management layer, confirming that the vulnerability is complete and uniform: every public-key operation is simultaneously broken by Shor’s algorithm. From this inventory we derive five formal threat scenarios with explicit adversarial assumptions and prioritise them in a severity–exploitability risk matrix. We show that the append-only ledger permanently amplifies the harvest-now-decrypt-later (HNDL) threat, and we adapt Mosca’s inequality to this setting across five deployment scenarios. The analysis identifies root-CA self-signing and orderer block-signing as the highest-priority migration targets and exposes a partial migration vulnerability in which post-quantum transaction signing gives no protection while the CA remains classical. We conclude with eight formal integration requirements, traceable to the identified threats, that specify a quantum-resistant Fabric architecture, including the requirement that the live CA itself issue post-quantum certificates at runtime.

Index Terms—post-quantum cryptography, Hyperledger Fabric, permissioned blockchain, quantum threat model, harvest-now-decrypt-later

I. INTRODUCTION

Quantum computing introduces a cryptographic threat fundamentally different from classical vulnerabilities. While classical attacks exploit implementation errors or weak parameters, Shor’s algorithm renders algorithms based on the discrete logarithm or integer factorisation problem insecure on a cryptographically relevant quantum computer (CRQC) [1]. Once such a device exists, migration to post-quantum cryptography (PQC) becomes the only effective defence.

Hyperledger Fabric, widely deployed across finance, healthcare, supply chain, and government [2], relies on ECDSA and ECDH P-256 throughout its transaction lifecycle and

identity infrastructure. Because both depend on the ECDLP, a CRQC could recover private keys, forge signatures, and decrypt recorded TLS sessions.

The harvest-now-decrypt-later (HNDL) threat is especially severe in blockchains. Fabric’s immutable ledger permanently stores transaction signatures, block-header signatures, and certificate chains, leaving historical cryptographic material vulnerable even after migration. For long-lived data, the threat therefore applies to the entire ledger history, not only future records. Despite growing recognition of quantum threats to blockchains [3], [4], no prior work has produced a component-level quantum vulnerability audit for Fabric, and existing PQC integration proposals lack formal threat analysis [5]–[8]. This paper makes the following contributions:

- The first component-level quantum vulnerability audit of Hyperledger Fabric is conducted, identifying twenty-five vulnerable operations across the transaction lifecycle and identity infrastructure.
- Five formal threat scenarios are defined, with explicit mapping to the HNDL (Harvest Now, Decrypt Later) dimensions.
- A risk prioritization matrix is proposed to facilitate the assessment and mitigation of identified vulnerabilities.
- A Mosca inequality analysis is performed to evaluate the urgency of quantum threats for representative Hyperledger Fabric deployments.
- Eight integration requirements are outlined to define a quantum-resistant architecture for Hyperledger Fabric.

Collectively, these contributions bridge the gap between theoretical quantum risk and practical engineering, providing the first actionable blueprint for securing Hyperledger Fabric against both immediate HNDL threats and future cryptanalytic breakthroughs.

II. METHODOLOGY

This section defines the scope, target system, and analytical framework of the audit, and sets out the four-stage process by which each cryptographic operation is enumerated, mapped, classified, and rated. It also establishes the severity rubric used

throughout the paper, calibrated on the scope of compromise rather than the reversibility of the consequence.

A. Scope and Target System

The audit targets Hyperledger Fabric v2.5.12 and covers the full transaction lifecycle, including the client SDK, endorsing peers, Raft ordering service, Fabric CA, and MSP. Sources include Fabric documentation, the Blockchain Cryptographic Service Provider (BCCSP) interface, and configuration schemas. Hash functions and AES-256 are excluded because, although affected by Grover’s algorithm [9], they retain acceptable post-quantum security under current NIST guidance [10]. The audit therefore focuses on asymmetric operations vulnerable to Shor’s algorithm.

B. Analytical Framework

Existing frameworks such as CVE, OWASP Cryptographic Failures, and NIST SP 800-57 [10] do not address the structural quantum vulnerability of correctly implemented ECDSA systems. The analysis therefore combines structured threat modelling [11], [12] with quantum-specific classification informed by NIST PQC literature [13], [14].

C. Four-Stage Audit Process and Severity Rubric

The audit, summarised in Figure 1, proceeds in four stages: (1) enumerates lifecycle cryptographic operations; (2) maps each to its algorithm and key type; (3) classifies operations as Shor-threatened or Grover-affected; and (4) assesses severity and exploitability. Severity is based on compromise scope. **Critical** applies to compromise of network-wide trust or trust anchors; **High** to transaction or identity-level compromise; **Medium** to confidentiality breaches such as TLS decryption; and **Low** to limited information exposure, though no operations in this audit fall into that category.

III. CRYPTOGRAPHIC INVENTORY

This section applies the audit process to Hyperledger Fabric v2.5.12, enumerating all identified 25 quantum-vulnerable operations across the five transaction lifecycle phases, the identity infrastructure, the gossip protocol, and the key management layer. It establishes the uniform ECDLP dependence of the platform and analyses the performance significance of signature verification at scale.

A. Uniform ECDLP Dependence

The audit reveals complete and uniform dependence on the ECDLP across all public-key functions in the standard lifecycle. Every digital signature uses ECDSA P-256; every session-key exchange uses ECDH P-256. There is no cryptographic diversity: all asymmetric guarantees rest on the same hardness assumption and are simultaneously eliminated by Shor’s algorithm. The audit identifies twenty-five distinct quantum-vulnerable operations across five lifecycle phases, two identity subsystems, the gossip protocol, and the key management layer (Table I).

Given an elliptic curve E over $\mathbb{F}_p$, a base point $G \in E(\mathbb{F}_p)$ of prime order n , and $Q \in E(\mathbb{F}_p)$, the ECDLP is to find the unique $k \in \{1, \dots, n-1\}$ with

$$Q = kG. \quad (1)$$

The best-known classical algorithm, Pollard’s rho, requires $O(\sqrt{n})$ group operations; for P-256, $n \approx 2^{256}$, giving $\approx 2^{128}$ operations. Shor’s algorithm reduces this to $O((\log n)^3)$ quantum gates [1]:

$$T_{\text{classical}} = O(\sqrt{n}), \quad T_{\text{quantum}} = O((\log n)^3). \quad (2)$$

Defining the security level λ as the binary logarithm of the best-known attack cost, the classical-to-quantum transition for ECDSA P-256 is

$$\lambda_{\text{classical}} \approx 128 \text{ bits}, \quad \lambda_{\text{quantum}} = 0. \quad (3)$$

Equation (3) establishes that no implementation or parameter change restores security once Shor’s algorithm is executable at the target key size — the property distinguishing the quantum vulnerability from all classical vulnerabilities.

B. Lifecycle Operations (Op-01–Op-16)

Phase 1 - Proposal/client signing. Op-01 (client proposal signing, ECDSA) authenticates client actions; forgery enables arbitrary proposals. High. Op-02 (client→peer TLS, ECDH) allows retroactive decryption of recorded traffic. Medium.

Phase 2 - Endorsement. Op-03 (proposal signature verification) protects transaction processing. High. Op-04 (X.509 chain validation) establishes identity authenticity; forged certificates enable channel-wide impersonation. Critical. Op-05 (endorsement signing) binds peers to transaction results; forgery enables fake endorsements. High. Op-06 (peer TLS) exposes long-term TLS keys. Medium.

Phase 3 - Assembly/submission. Op-07 (envelope signing) authenticates endorsed transactions. High. Op-08 (client→orderer TLS) exposes pending transactions before commitment. Medium.

Phase 4 - Ordering/block signing. Op-09 (envelope verification) is the final authentication step before ordering. High. Op-10 (block-header signing) is the most critical operation, since forged signatures enable irreversible insertion of fraudulent blocks. Critical. Op-11 (orderer→peer TLS) protects block delivery. Medium. Op-12 (Raft mutual TLS authentication) protects consensus integrity. High. Op-13 (Raft TLS key exchange) exposes consensus traffic and may enable manipulation when combined with Op-12. High.

Phase 5 - Committing-peer validation. Op-14 (endorsement re-verification) is the most performance-sensitive operation under PQC and creates permanent HNDL targets. High. Op-15 (block-header verification) is the final integrity check before commit. Critical. Op-16 (certificate re-validation) is the final defence against fraudulent certificates before permanent ledger commitment. Critical.

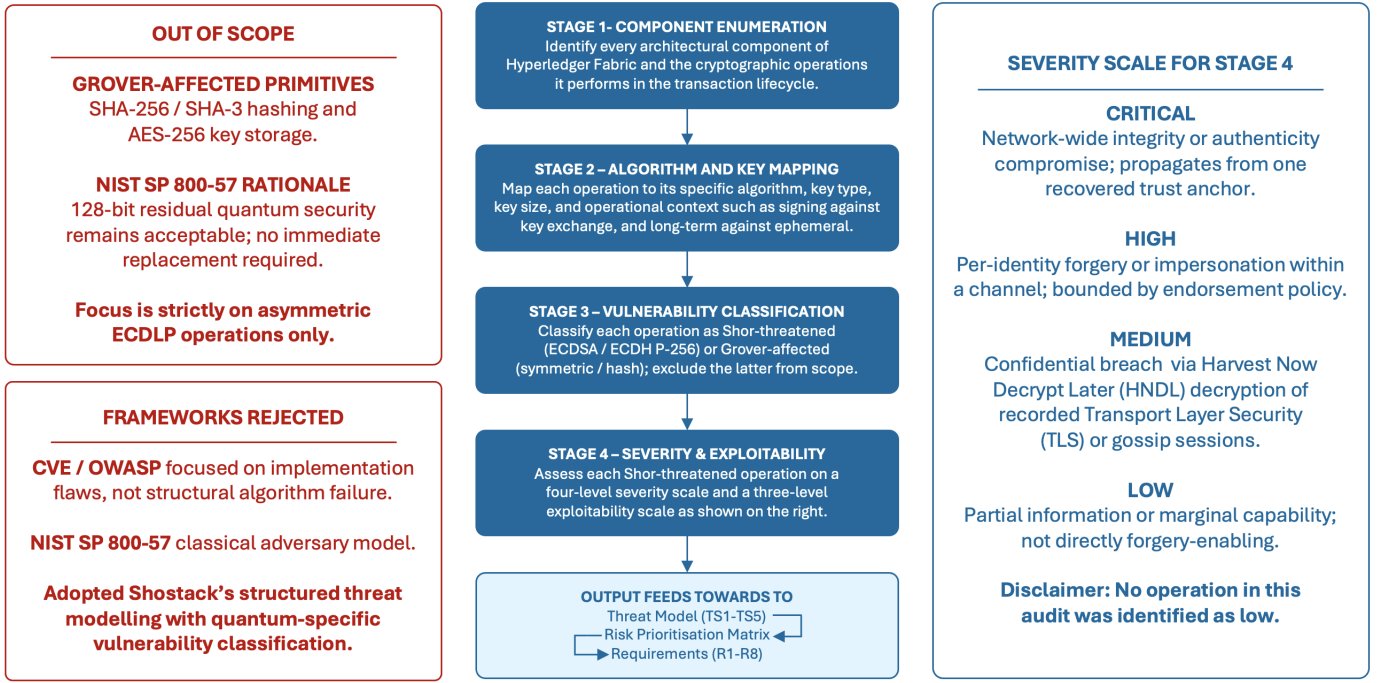


Fig. 1: Hyperledger Fabric Quantum Vulnerability Audit Methodology

C. Identity Infrastructure, Gossip, and Key Management (Op-17–Op-25)

Fabric CA. Op-17 (root CA self-signing) is the network trust anchor; compromise enables arbitrary certificate issuance. Critical. Op-18 (intermediate CA signing) extends compromise through the CA hierarchy. Critical. Op-19 (end-entity issuance) affects identities under the compromised CA. High. Op-20 (CA enrolment TLS) exposes participant identity data. Medium.

MSP. Op-21 (MSP validation) accepts certificates from a compromised root CA. Critical. Op-22 (channel configuration signing) forgery enables arbitrary policy and organisation changes. Critical.

Gossip and key management. Op-23 (gossip signing) can disrupt peer synchronisation. Medium. Op-24 (gossip TLS) exposes uncommitted data. Medium. Op-25 (BCCSP keypair generation) requires full identity regeneration during PQC migration. High.

D. Signature Verification at Scale

Op-14 (endorsement re-verification) is executed by every committing peer for every transaction, processing thousands of verifications per second at moderate throughput. The performance impact of Post-Quantum Cryptography (PQC) is amplified beyond verification time: larger signatures increase block sizes, slowing gossip propagation and reducing sustainable throughput. Since each transaction carries multiple endorsements with full certificate chains, replacing compact ECDSA signatures with significantly larger PQC alternatives severely impacts storage and network bandwidth. Consequently, *signature*

output size is a more critical performance determinant than signing speed for blockchain PQC migration.

Let T_b be transactions per block, $\bar{e}$ the average endorsers per transaction, $|\sigma_{\text{alg}}|$ the signature size, and $|cert_{\text{alg}}|$ the certificate-chain size. The Op-14 cryptographic payload per block is

$$B_\sigma = T_b \cdot \bar{e} \cdot (|\sigma_{\text{alg}}| + |cert_{\text{alg}}|). \quad (4)$$

With $\tau(|\sigma|)$ the per-signature verification time and C_{val} a peer's validation capacity, the maximum sustainable throughput is bounded by

$$\Gamma_{\text{max}} \leq \frac{C_{\text{val}}}{T_b \cdot \bar{e} \cdot \tau(|\sigma_{\text{alg}}|)}. \quad (5)$$

A smaller $|\sigma_{\text{alg}}|$ reduces B_σ in (4), decreasing propagation latency and relaxing the bound in (5) even when $\tau(|\sigma|)$ is higher — motivating FN-DSA-512 for throughput-sensitive deployments despite its slower signing relative to ML-DSA-44.

E. Complete Inventory

Figure 2 maps all twenty-five quantum-vulnerable operations across eight component groups: the five transaction-lifecycle phases, the Fabric CA/MSP identity infrastructure, the gossip protocol, and the key management layer. Fabric CA/MSP contains the highest concentration of Critical operations, with Op-17 forming the primary TS1 attack surface because the root CA key remains exposed in the genesis block. ECDH-based TLS operations (Op-02, 06, 08, 11, 13, 20, 24) appear throughout the lifecycle, showing that HNDL affects all communication layers.

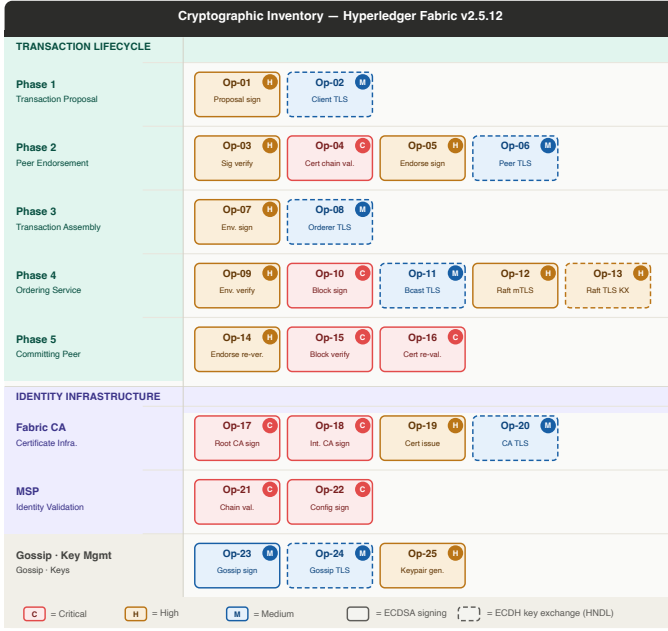


Fig. 2: Cryptographic inventory of Hyperledger Fabric v2.5.12, showing all twenty-five quantum-vulnerable operations.

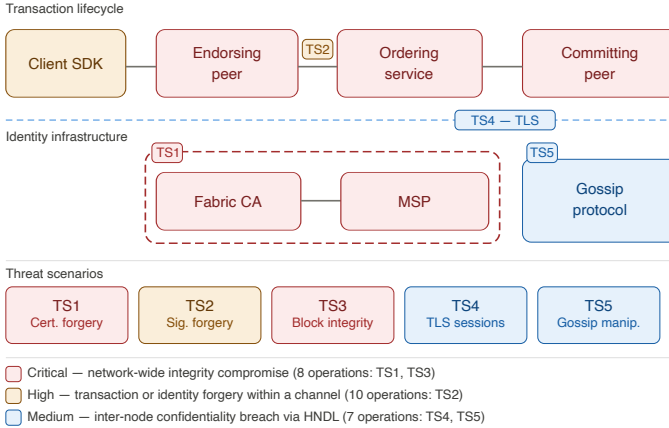


Fig. 3: Quantum vulnerability overview for Hyperledger Fabric v2.5.12, color-coded by severity across six components.

Figure 3 summarises the component-level threat model. Red components contain Critical operations, while the dashed red overlay marks the TS1 attack surface. The dashed blue layer represents TLS-based TS4 exposure across inter-component communication.

Table I lists all twenty-five operations with algorithm and severity. The vulnerability is complete (every public-key operation) and uniform (every operation ECDLP-based). The distribution is eight critical, ten high, and seven medium, with critical operations concentrated in the certificate infrastructure, the block signing/verification path, and MSP validation confirming that the integrity backbone is most directly threatened.

TABLE I: Complete cryptographic inventory of Hyperledger Fabric v2.5.12. All twenty-five operations use ECDLP-based mathematics and are threatened by Shor’s algorithm.

Op	Operation	Algorithm	Severity
Phase 1 : Transaction Proposal			
01	Client transaction proposal signing	ECDSA P-256	High
02	Client TLS handshake to peer	ECDH P-256	Medium
Phase 2 : Peer Endorsement			
03	Client signature verification at peer	ECDSA P-256	High
04	X.509 certificate chain validation	ECDSA P-256	Critical
05	Chaincode execution result signing	ECDSA P-256	High
06	Peer TLS inbound/outbound sessions	ECDH P-256	Medium
Phase 3 : Transaction Assembly			
07	Client transaction envelope signing	ECDSA P-256	High
08	Client-to-orderer TLS session	ECDH P-256	Medium
Phase 4 : Ordering Service			
09	Envelope sig. verification at orderer	ECDSA P-256	High
10	Block header signing by orderer	ECDSA P-256	Critical
11	Orderer-to-peer TLS (block broadcast)	ECDH P-256	Medium
12	Raft inter-orderer mutual TLS auth	ECDSA P-256	High
13	Raft inter-orderer TLS key exchange	ECDH P-256	High
Phase 5 : Committing Peer Validation			
14	Endorsement sig. re-verification	ECDSA P-256	High
15	Block header signature verification	ECDSA P-256	Critical
16	Certificate chain re-validation	ECDSA P-256	Critical
Fabric CA: Certificate Infrastructure			
17	Root CA certificate self-signing	ECDSA P-256	Critical
18	Intermediate CA certificate signing	ECDSA P-256	Critical
19	End-entity certificate issuance	ECDSA P-256	High
20	Fabric CA TLS session	ECDH P-256	Medium
MSP: Identity Validation			
21	MSP certificate chain validation	ECDSA P-256	Critical
22	MSP channel configuration signing	ECDSA P-256	Critical
Gossip Protocol			
23	Gossip message signing	ECDSA P-256	Medium
24	Gossip TLS sessions	ECDH P-256	Medium
Key Management			
25	BCCSP keypair generation	ECDSA P-256	High

IV. QUANTUM THREAT MODEL

This section translates the inventory into five formal threat scenarios, each with explicit adversarial capability assumptions, impact assessment, and a harvest-now-decrypt-later dimension specific to the immutable ledger. The scenarios range from network-wide certificate forgery to confidentiality-only TLS decryption.

A. Adversarial Capability Model

The adversary possesses a CRQC capable of executing Shor’s algorithm against P-256; breaking ECDSA P-256 requires approximately 2,330 logical qubits with error correction [15], with similar estimates for RSA-2048 [16]. Although such devices do not yet exist, HNDL makes them relevant for long-term data [17]. Classically, the adversary can observe

network traffic and access public ledger, configuration, and historical transaction data without privileged access. Physical node compromise and classical Raft-majority corruption are excluded.

B. TS1: Identity Impersonation via Certificate Forgery

Severity: Critical. Exploitability: High. Priority: P1. TS1 targets the root CA certificate in the genesis block. Using publicly available MSP configurations and a CRQC, an adversary applies Shor’s algorithm to recover the ECDSA root private key and issue valid certificates for arbitrary identities and roles. Forged administrator signatures also enable fraudulent channel updates (Op-22). Because trust propagates through the CA hierarchy, compromising the root key compromises all derived certificates. The HNDL risk is permanent since the root key remains embedded on-chain from genesis. *Affected operations: Op-04, Op-16, Op-17, Op-18, Op-19, Op-21, Op-22.*

C. TS2: Transaction Signature Forgery

Severity: High. Exploitability: High. Priority: P2. TS2 targets ECDSA signatures on proposals, endorsements, and transaction envelopes. Since transaction records expose verification keys, any ledger reader can harvest them. After recovering a private key with Shor’s algorithm, the adversary can sign fraudulent transactions or endorsements as the compromised identity, including bypassing multi-organisation endorsement policies. The HNDL threat persists permanently because historical ledger data cannot be removed. *Affected operations: Op-01, Op-03, Op-05, Op-07, Op-09, Op-14.*

D. TS3: Block Integrity Compromise via Orderer Key Recovery

Severity: Critical. Exploitability: High. Priority: P1. TS3 targets orderer block-signing keys exposed in block headers. Recovering an orderer private key enables valid signatures on fraudulent blocks, allowing insertion, reordering, or suppression of transactions. Since committed blocks become permanently embedded in the ledger hash chain, integrity compromise is effectively irreversible. Compromising Raft inter-orderer authentication (Op-12/Op-13) further enables consensus manipulation through MITM attacks. *Affected operations: Op-10, Op-12, Op-13, Op-15.*

E. TS4: TLS Session Decryption via HNDL

Severity: Medium. Exploitability: Medium. Priority: P3. TS4 targets ECDH P-256 used in TLS sessions. An adversary records encrypted traffic and later uses Shor’s algorithm to recover session keys from captured handshakes, decrypting historical communications. Exposed data may include proposals, endorsements, block contents, and administrative traffic. Ephemeral ECDH provides no effective forward secrecy against a quantum adversary. *Affected operations: Op-02, Op-06, Op-08, Op-11, Op-13, Op-20, Op-24.*

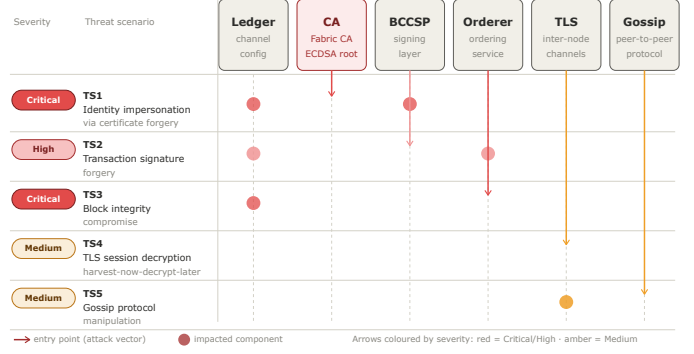


Fig. 4: Threat scenario attack paths mapped to Fabric components: entry points and impacted components.

F. TS5: Gossip Protocol Manipulation

Severity: Medium. Exploitability: Low. Priority: P4. TS5 recovers a peer’s gossip signing key from observable discovery messages and injects fraudulent gossip, or decrypts recorded gossip TLS. Two impacts arise: state-synchronisation manipulation (false block announcements, unnecessary state transfers, incorrect ledger height, or restoring a recovering peer from a fraudulent state) and transaction-content exposure (advance knowledge of propagating block content, constituting material non-public information in financial networks). Exploitability is Low because TS5 requires a network position to observe or inject gossip, unlike the P1/P2 threats executable by any ledger reader. *Affected operations: Op-23, Op-24.*

Figure 4 maps each scenario to its entry point (downward arrow) and impacted components (filled circles). TS1 and TS3 originate from key material permanently accessible on the ledger (genesis-block CA certificate; orderer key in block headers); TS2 originates at the BCCSP signing layer; TS4/TS5 require active network observation and are rated Medium.

V. RISK PRIORITISATION

This section orders the five threat scenarios by severity and exploitability in a risk matrix, derives the resulting migration priorities, and formalises the analysis with a composite risk score. It also exposes the partial migration vulnerability, in which post-quantum transaction signing provides no protection while the certificate authority remains classical.

A. Risk Matrix

Table II assesses each scenario by severity and exploitability, determining migration priority. TS1 and TS3 share the joint highest priority (Critical, High); TS2 is second (High, High); TS4 and TS5 are lower, constrained by network-observation requirements more demanding than passive ledger reading.

Three architectural conclusions follow. First, the CA infrastructure and the block-signing path share top priority and must be addressed simultaneously: both are accessible to any passive ledger observer, and no measure short of replacing both with PQC addresses the P1 threats. Second, peer endorsement and client signing share second priority — more limited in scope but equally exploitable, since signatures and keys appear

TABLE II: Risk prioritization matrix for the five quantum threat scenarios identified in the Hyperledger Fabric v2.5.12 audit.

Prio.	Scenario	Severity	Exploit
P1	TS1: Identity Impersonation	Critical	High
P1	TS3: Block Integrity Compromise	Critical	High
P2	TS2: Tx Signature Forgery	High	High
P3	TS4: TLS Session Decryption	Medium	Medium
P4	TS5: Gossip Manipulation	Medium	Low

in every historical record. Third, TLS and gossip occupy lower priority only because exploitation requires network observation; they remain in the requirements because complete quantum resistance requires the full attack surface, and HNDL on TLS requires only past observation.

B. Quantitative Risk Synthesis

Mapping severity to $s_i \in \{1, 2, 3\}$ (M/H/C) and exploitability to $e_i \in \{1, 2, 3\}$ (L/M/H), the composite score is

$$\rho_i = s_i \cdot e_i. \quad (6)$$

TS1 and TS3 yield $\rho = 9$ (P1, maximum); TS2 yields 6 (P2); TS4 yields 2 (P3); TS5 yields 1 (P4). No measure beneath the cryptographic layer can reduce s_i or e_i for TS1/TS3, as their exploitability derives from immutable ledger content. Defining the coverage relation $\Phi \subseteq \{R_1, \dots, R_8\} \times \{TS1, \dots, TS5\}$ with $(R_k, TS_j) \in \Phi$ iff R_k addresses TS_j , the requirement set is complete iff

$$\forall j \in \{1, \dots, 5\}, \exists k \in \{1, \dots, 8\} : (R_k, TS_j) \in \Phi. \quad (7)$$

Table IV confirms (7) holds (TS1 by R_3 , TS2/TS3 by R_1 , TS4/TS5 by R_2); completeness is bounded by the scope of the threat enumeration. Finally, with $\Gamma(t)$ the throughput, $\bar{k}$ the mean new ECDSA keys per transaction, and $\mathbb{1}[t < t_m]$ the pre-migration indicator, the harvest set grows at

$$\frac{d|\mathcal{H}|}{dt}(t) = \Gamma(t) \cdot \bar{k} \cdot \mathbb{1}[t < t_m], \quad (8)$$

so total accumulated exposure from genesis t_0 to evaluation t_{eval} is

$$|\mathcal{H}(t_{\text{eval}})| = \int_{t_0}^{\min(t_m, t_{\text{eval}})} \Gamma(t) \bar{k} dt \approx \bar{\Gamma} \bar{k} (t_{\text{eval}} - t_0). \quad (9)$$

Equation (9) makes precise the governance conclusion: every day under ECDSA P-256 permanently enlarges the harvest set by $\approx \bar{\Gamma} \cdot \bar{k} \cdot 86,400$ keys.

C. The Partial Migration Vulnerability

A critical insight is the structural interdependence between the CA layer and the BCCSP signing layer. If an operator deploys PQC signing at the BCCSP layer without migrating the certificate infrastructure, the CA public keys in the channel configuration remain ECDSA. A quantum adversary who

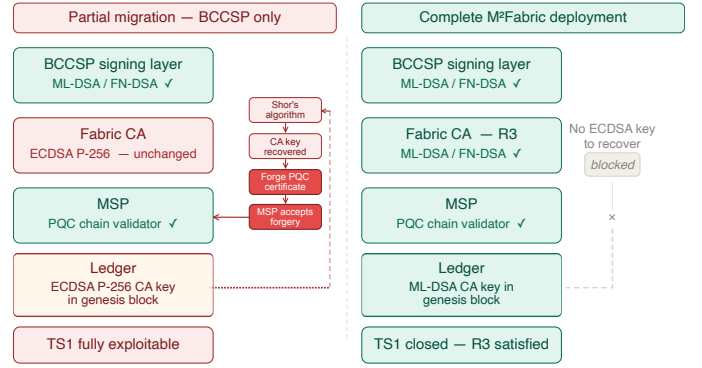


Fig. 5: Partial migration vulnerability: TS1 remains fully exploitable when only the BCCSP layer is migrated.

recovers the CA private key issues fraudulent *post-quantum* certificates accepted as legitimate by any PQC-capable MSP — the quantum resistance of the signing layer is completely neutralised by the classical vulnerability of the certificate layer.

Figure 5 contrasts partial migration (left, red=vulnerable) with full compliance (right, teal=secure). In the left panel, although BCCSP/MSP are migrated, the legacy ECDSA CA key in the genesis block allows an attacker to use Shor’s algorithm to forge valid post-quantum certificates, which the MSP inadvertently accepts. This demonstrates that migrating signing algorithms alone is insufficient if the CA remains classical. In the right panel, satisfying R3 by issuing post-quantum CA keys eliminates the harvestable ECDSA material, thereby closing TS1. Thus, R3 is strictly required; deferring CA migration leaves the system vulnerable regardless of BCCSP upgrades.

VI. QUANTUM ATTACK TIMELINE ANALYSIS

This section establishes the urgency of migration by relating the security lifetime of protected data to the projected CRQC emergence date and the migration lead time. It applies Mosca’s inequality across five representative Fabric deployment scenarios and derives a blockchain-corrected formulation that accounts for the permanent harvest exposure of historical ledger records.

A. Mosca’s Inequality

Urgency requires a temporal analysis relating data security lifetime to the CRQC emergence date and migration lead time. Mosca’s inequality states that if $X + Y > Z$ migration must begin immediately, where X is years until a CRQC, Y is migration duration, and Z is the required data security lifetime [17]:

$$X + Y > Z \implies \text{PQC migration must begin immediately.} \quad (10)$$

Anchored to a 2026 planning base, the optimistic estimate places $X = 7$ years (CRQC ≈ 2033), informed by Mosca’s assessment of a one-in-two chance by 2031 [17]; the pessimistic estimate places $X = 15$ years (≈ 2041).

Migration time Y for an enterprise Fabric network is estimated at 2–3 years, accounting for a production-quality PQC release, certificate-infrastructure renewal for all identities, staged multi-organisation rollout, and operational/regulatory validation — consistent with major financial-sector PQC planning.

B. Application to Fabric Deployment Scenarios

Table III evaluates the inequality across five scenarios under both timelines. Three of five satisfy $X + Y > Z$ under both estimates, indicating immediate action regardless of CRQC timing: healthcare (30+ year retention), financial settlement (20+ years), and government procurement (25+ years). Supply chain is *Borderline/Immediate*: records span $Z \approx 5$ (inventory) to $Z \approx 10$ (regulated provenance); under the optimistic timeline ($X + Y = 10$) the inequality holds at the low end but not strictly at $Z = 10$, while under the pessimistic timeline ($X + Y = 18$) it holds throughout. Only data with $Z \leq 2$ years falls safely below the horizon under both timelines.

TABLE III: Application of Mosca’s inequality to five representative Hyperledger Fabric deployment scenarios, under optimistic and pessimistic CRQC timelines.

Scenario	Z (yr)	X opt	X pess	Y	Action
Healthcare records	30+	7	15	3	Immediate
Financial settlement	20+	7	15	3	Immediate
Govt. procurement	25+	7	15	3	Immediate
Supply chain	5–10	7	15	3	Borderline/Imm.
Two-year contracts	2	7	15	3	Can defer

C. HNDL Amplification in Blockchain Deployments

The conventional inequality evaluates Z against the desired future lifetime; for blockchains this is insufficient, since every previously committed record is also a harvest target. With t_0 genesis, t_m migration completion, and $\mathcal{L}(t)$ the keys committed up to t , the HNDL exposure set is

$$\mathcal{H}(t) = \{ \text{pk}_i \in \mathcal{L}(t) \mid \text{pk}_i \text{ generated under ECDSA P-256} \}, \quad (11)$$

and because the ledger is append-only, $|\mathcal{H}(t)|$ is non-decreasing. Migration at t_m prevents new additions but cannot reduce $|\mathcal{H}(t_m)|$, which remains permanently positive. The blockchain-corrected parameter substituted into (10) is

$$Z_{\text{blockchain}} = Z_{\text{future}} + (t_{\text{eval}} - t_0) \geq Z_{\text{future}}. \quad (12)$$

For a network operational since 2018 and evaluated in 2025, $(t_{\text{eval}} - t_0) = 7$ years of records are permanently in $\mathcal{H}$, extending the effective Z by 7 years regardless of when migration completes. Migration should therefore be completed as early as possible to minimise additional vulnerable records, while affected organisations must accept some permanent residual exposure for historical records.

VII. FORMAL INTEGRATION REQUIREMENTS

The threat analysis establishes eight integration requirements for a complete quantum-resistant Fabric architecture.

R1: PQC signing at lifecycle signing points. Addresses TS2/TS3 across operations including transactions, endorsements, blocks, and configuration updates. Approved algorithms include ML-DSA (FIPS 204) and FN-DSA (FIPS 206) [18]; SLH-DSA (FIPS 205) is unsuitable for transaction use due to large signatures.

R2: PQC key encapsulation for TLS. Addresses TS4 and TLS-related TS5 operations using ML-KEM-768 (FIPS 203). During migration, hybrid ECDH+ML-KEM TLS is required [19].

R3: Runtime PQC certificate issuance by the Fabric CA. Addresses TS1. The live CA must issue PQC X.509 certificates, since an ECDSA CA leaves newly enrolled identities vulnerable.

R4: Preserve the BCCSP interface. PQC integration must remain confined to a BCCSP plugin without changes to higher Fabric layers.

R5: Crypto-agility through configuration. Algorithms must be configurable rather than hardcoded to support future migration needs.

R6: Functional correctness. PQC transactions must preserve endorsement, validation, execution, and commitment behaviour.

R7: Constant-time PQC implementations. Signing and key-generation operations must resist timing attacks, particularly for FN-DSA [20].

R8: Phased migration support. The architecture must support incremental deployment across organisations without requiring simultaneous upgrades.

TABLE IV: Requirements traceability matrix for the eight formal integration requirements derived from the quantum threat analysis.

Req.	Description	Addresses	Type
R1	PQC signing at all signing points	TS2, TS3	Functional
R2	PQC key encapsulation in TLS	TS4, TS5	Functional
R3	PQC certificates from live CA	TS1	Functional
R4	Preserve BCCSP interface	Quality constraint	Architectural
R5	Crypto-agility via configuration	Quality constraint	Architectural
R6	Functional correctness	Quality constraint	Quality
R7	Constant-time operations	Side-channel	Security
R8	Phased migration pathway	Operational	Operational

Table IV confirms two properties. First, every threat is addressed by at least one functional requirement (TS1→R3, TS2/TS3→R1, TS4/TS5→R2), so the set is complete. Second, the traceability of R3 to TS1 alone formalises the partial migration vulnerability: no subset omitting R3 can address the full model. The non-functional requirements R4–R8 are necessary for a deployable, maintainable, and secure implementation.

VIII. DISCUSSION

This section discusses the completeness of the threat model, the amplification of the harvest-now-decrypt-later (HNDL)

threat by ledger immutability, and the generalisability of the methodology to other permissioned blockchain platforms.

Completeness. The inventory identifies a broader quantum attack surface than prior work by including Raft consensus, gossip signing/encryption, and MSP channel-configuration signing alongside transaction-signing and CA layers. Campbell [6] discussed the PKI threat without operational enumeration, while Holcomb et al. [5], Hu [7], and Abbas et al. [21] implemented solutions without formal threat models. The inventory also enables evaluation of these approaches: Holcomb et al. address Op-05 and partially Op-14/Op-15 but leave TS1 (Op-17–Op-22) unresolved, while Abbas et al. mitigate Op-19 through offline generation but leave the live CA exposed. Because all cryptographic components rely on ECDLP, Shor’s algorithm compromises the entire architecture without protection from algorithm diversity.

Ledger immutability amplification. A key finding is that immutable ledgers permanently amplify the HNDL threat. Unlike conventional PKI, where migration and key rotation can reduce exposure, historical ECDSA public keys stored on-chain remain permanent harvest targets, including those of removed participants. HNDL risk therefore accumulates continuously from genesis and cannot be eliminated retrospectively through migration alone.

Generalisability. Although the analysed operations are Fabric-specific, the methodology, threat scenarios, and requirements extend to other permissioned blockchain platforms such as Besu, Quorum, and Corda, which also rely on certificate-based ECDSA and TLS infrastructure. The identified threat categories identity impersonation, transaction-signature forgery, node-key compromise, TLS decryption, and peer-protocol manipulation recur across permissioned blockchain systems, as does the need to separate certificate-infrastructure migration from signing-layer migration while supporting runtime PQC certificate issuance.

IX. CONCLUSION

This paper presented the first component-level quantum vulnerability audit and formal threat model for Hyperledger Fabric, identifying twenty-five ECDLP-based operations vulnerable to Shor’s algorithm. Block header signing (Op-10) and Root CA self-signing (Op-17) are the most critical due to permanent Harvest Now, Decrypt Later (HNDL) exposure, while endorsement re-verification (Op-14) poses the greatest performance challenge. The highest-priority threats are HNDL attacks on ledger history (TS1) and CA infrastructure compromise (TS3). Blockchain-corrected Mosca analysis confirms urgent migration needs for healthcare, financial, and government sectors. Crucially, mitigating TS1 requires a live Fabric CA capable of runtime post-quantum certificate issuance (R3). These findings define a complete quantum-resistant architecture, providing a definitive roadmap for securing enterprise blockchain infrastructure against imminent quantum threats.

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